Replication-induced instability of a certified five-cluster motor cycle

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Abstract

An exactly repeated collective motion need not retain its stability when its constituent groups are enlarged. We give a computer-assisted example in the first-harmonic molecular-motor model of Alouges and collaborators. At explicit parameters, an eight-row system has an attracting periodic orbit with five distinct cluster states and multiplicities $2+2+1+2+1$. Interval integration, multiple shooting and spectral tests certify its existence and place all 22 nonflow Floquet multipliers inside modulus 0.13. Exact replication to $N = 8k$ rows preserves the orbit and the cluster weights, but gives precisely $2k - 2$ unstable, one neutral and $22k$ stable directions. The instability is carried by internal differences in the two groups that were singletons at $N = 8$. Each newly available block has one real multiplier in $(-23.084, -23.083)$ and two multipliers of modulus below 0.13. The proof combines an explicit invariant splitting with reproducible interval certificates. Full-system numerical trajectories illustrate attraction and sign-reversing departure without serving as premises of the theorem.

Keywords: cluster synchronization; periodic orbit; Floquet stability; molecular motors; validated numerics.

1 Introduction

Collective oscillations are often represented by a small collection of distinct group states. Such a description retains the motion within the cluster subspace but can hide perturbations that split an individual group. This distinction becomes consequential when a small configuration is replicated: the same group trajectories may survive exactly while new internal degrees of freedom make them unstable.

Alouges et al. [1] introduced a model for interactions between rows of molecular motors in an axonemal cross-section. Their eight-row numerical examples display periodic motion organized into two groups; the organization of the general $3N - 1$ dimensional system is left open. We study an explicit five-cluster cycle in its exact first-harmonic specialization. The question is whether attraction of this motion survives balanced enlargement of every cluster.

Symmetry-based separation of cluster motion and transverse perturbations is established in synchronization theory [2, 3]. Connections between different cluster patterns, symmetry breaking and chaos have also been studied in globally coupled oscillators [4]. Our contribution is an explicitly enclosed orbit and a complete stability classification of its replicated family in the motor equations. The proof resolves every tangent direction, including ones absent in the smallest member of the family.

The central result is that the $N = 8$ cycle attracts locally, whereas all replicas with $N = 8k$, $k \geq 2$, are saddles. The instability is strong: a perturbation in each new unstable mode reverses sign and grows in magnitude by about 23.08 over a certified return period. This is a change in the available perturbation space as the system size changes. The theorem concerns fixed

normalized parameters and fixed cluster weights. It makes no identification of this enlargement with changing the anatomy of a physical flagellum while holding all dimensional constants fixed.

2 Model and main result

For $N \geq 2$, write the state of row j as (c_j, s_j, X_j) and impose $\sum_j X_j = 0$. In normalized time τ, the equations are

$$V_j = -\rho X_j + r(s_j - \bar{s}), \qquad \bar{s} = \frac{1}{N}\sum_{i=1}^{N} s_i, \tag{1}$$

$$c_j' = 1 - c_j - V_j s_j, \qquad s_j' = b - s_j + V_j c_j, \qquad X_j' = V_j. \tag{2}$$

The first harmonic of the motor density determines (c_j, s_j); X_j is the normalized relative displacement. Eliminating X_N gives a $3N - 1$ dimensional autonomous polynomial system. The mean term in (1) follows from the mechanical displacement constraint. The normalization and its relation to the published equations are given in the accompanying source audit. This manuscript uses

$$\rho_* = \frac{129}{55}, \qquad r_* = 1 + \frac{19/20}{\sqrt{32001/500000}}, \qquad b_* = 1. \tag{3}$$

The value of r_* agrees with the activity path implemented from the source model. The rational value of ρ_* specifies the theorem exactly and avoids identifying a floating-point grid point with an exact parameter.

Theorem 1 (Certified cycle and complete replication index)**.** *At* (3)*, the $N = 8$ system has a nonconstant periodic orbit with five pairwise distinct cluster states and multiplicities $(m_1, \ldots, m_5) = (2, 2, 1, 2, 1)$. It has a positive return period satisfying*

$$25.0016843758970971 < T < 25.0016843758970972. \tag{4}$$

For every integer $k \geq 1$, replication to multiplicities $(2k, 2k, k, 2k, k)$ is an exact periodic solution of the $N = 8k$ system at the same parameters and with the same return period. Its Floquet index is

$$(u, n, s) = (2k - 2,\ 1,\ 22k), \tag{5}$$

where u, n, s count multipliers outside, on and inside the unit circle, with algebraic multiplicity. The sole neutral multiplier is the simple flow multiplier 1. Every stable multiplier has modulus below 0.13. Every unstable multiplier belongs to $(-23.084, -23.083)$. In particular, the $N = 8$ orbit is locally orbitally exponentially attracting; the $N = 8k$ replicas are hyperbolic periodic saddles for $k \geq 2$.

The theorem certifies a return period. It does not assert that the displayed T is the least positive period. The stability and index conclusions do not require such an assertion: taking a positive integer power of a minimal-period monodromy preserves the numbers of stable and unstable directions. The simplicity of the flow multiplier is certified for the displayed return.

3 Invariant splitting and replication

Partition the rows into K nonempty groups of sizes m_a and write $w_a = m_a/N$. On the cluster subspace, all rows of group a have the same state (c_a, s_a, X_a). Substitution in (1)–(2) gives the weighted quotient

$$V_a = -\rho X_a + r\left(s_a - \sum_{b=1}^{K} w_b s_b\right), \qquad \sum_{a=1}^{K} w_a X_a = 0, \tag{6}$$

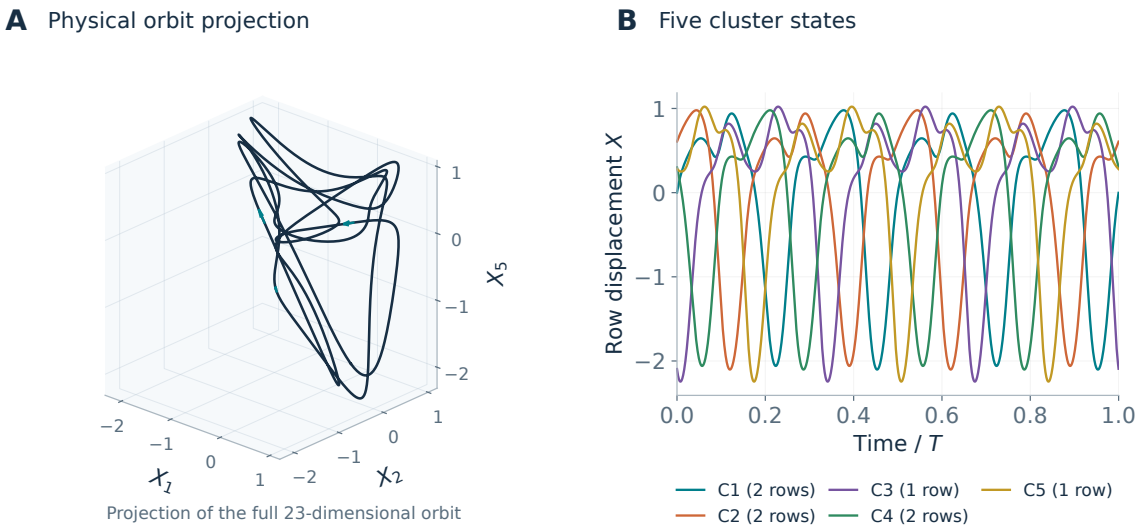


Figure 1. The certified five-cluster orbit. (A) A numerical rendering in three physical displacement coordinates; the full state has 23 independent coordinates. Arrows indicate increasing time. (B) One row displacement from each distinct cluster over the certified return period, with cluster sizes in the legend. Geometric closure in panel A is illustrative; existence follows from interval multiple shooting.

with the same three row equations. It has dimension $3K-1$. Multiplying every m_a by the same positive integer leaves all weights unchanged. Therefore every quotient trajectory, periodic or otherwise, embeds exactly in every balanced replica.

Along a cluster trajectory, an internal perturbation has zero sum within one cluster in each of its three coordinates. It changes neither the global mean of s nor the displacement constraint. Each independent internal contrast in group a obeys

$$\xi' = A_a(\tau)\xi, \qquad A_a = \begin{pmatrix} -1 & -V_a - r s_a & \rho s_a \\ V_a & -1 + r c_a & -\rho c_a \\ 0 & r & -\rho \end{pmatrix}. \tag{7}$$

There are $m_a - 1$ such three-dimensional copies. The remaining tangent directions are constant within each cluster and form the quotient tangent space. This is a direct sum: decomposing row perturbations into their cluster averages and zero-sum residuals gives unique components.

Proposition 2 (Replication spectrum)**.** *Let M_Q be the quotient monodromy over a positive period T, and let M_a be the return matrix of (7), whether or not the group is a singleton in the original system. The monodromy of the replica with multiplicities km_a is similar to*

$$M_Q \;\oplus\; \bigoplus_{a=1}^{K} \left(\mathrm{I}_{km_a-1} \otimes M_a\right). \tag{8}$$

Equivalently, its characteristic polynomial is

$$\chi_{8k}(z) = \chi_Q(z) \prod_{a=1}^{5} \chi_a(z)^{km_a-1} \tag{9}$$

for the five-cluster family considered here.

Proof. The cluster-average and zero-sum spaces described above are invariant under the variational equation. On each contrast copy the equation is (7); on the average space it is the

derivative of (6). The dimensions sum to $(3K-1)+3\sum_a(km_a-1)=3kN-1$, so no direction is omitted. Integration over T preserves the direct sum and gives (8). The determinant factorization yields (9). □

For a singleton, $m_a-1=0$. Its matrix M_a describes a mathematically well-defined response along the orbit, but contributes no direction to the original tangent space. Replication makes this matrix dynamically relevant. Consequently attraction at $N=8$ puts no stability restriction on the two absent singleton blocks. This identifies precisely the information needed in addition to the original stability certificate.

4 Computer-assisted proof

The proof uses outward-enclosing Arb ball arithmetic through Python-FLINT 0.9.0 [5]. The inclusion argument is a finite-dimensional verification argument of the type surveyed in [6]. The custom flow integrator, complete inputs, lossless interval outputs and executed source snapshots are provided in the repository. Ordinary numerical shooting supplies initial proposals only.

4.1 Relative periodicity and validated shooting

Let q denote the 14 independent quotient coordinates, with the last displacement eliminated using (6). Define a weight-preserving permutation by

$$(Pq)_a = q_{\pi(a)}, \qquad \pi=(4,1,5,2,3), \qquad P^6=\mathrm{I}, \tag{10}$$

where the list specifies the image index for each group. The certified relative closure is $\phi_L(q_0)=Pq_0$ with $L=T/6$. It cyclically exchanges the three paired groups and swaps the singletons.

We divide L into three equal segments of duration $\ell=L/3$. Unknowns are q_0,q_1,q_2 and ℓ, with the phase fixed by $X_1(q_0)=0$. There are 42 free scalar unknowns and 42 equations:

$$F(q_0,q_1,q_2,\ell)=\begin{pmatrix}\phi_\ell(q_0)-q_1\\ \phi_\ell(q_1)-q_2\\ \phi_\ell(q_2)-Pq_0\end{pmatrix}=0. \tag{11}$$

The Jacobian includes the endpoint vector field as the derivative with respect to duration. A ball $\mathcal{B}$ around an approximate solution $\hat{q}$ and a fixed invertible preconditioner C satisfy

$$-CF(\hat{q})+\big(\mathrm{I}-C[DF(\mathcal{B})]\big)(\mathcal{B}-\hat{q})\subset\operatorname{int}(\mathcal{B}-\hat{q}). \tag{12}$$

The induced infinity norm of $\mathrm{I}-C[DF(\mathcal{B})]$ is strictly below one. Thus $q\mapsto q-CF(q)$ is a contraction of the box into itself; its unique fixed point is a zero of F because C is invertible. The verified positive section velocity and duration exclude a stationary solution. Pairwise disjoint coordinate enclosures at the section verify that all five cluster states are distinct. Uniqueness of the ODE then prevents any two cluster states from coinciding at another time.

Each flow step first encloses a Picard tube for all uncertain initial states and durations and checks a contraction bound. The polynomial Taylor recurrence is then evaluated at the input box; its next-order coefficient is enclosed over the entire tube to bound the remainder. The same procedure integrates the variational blocks. Bounds therefore include truncation, rounding and input uncertainty.

4.2 Full-space attraction at eight rows

Write $H = D\phi_L(q_0)$ and $Q = P^{-1}H$. Equivariance gives

$$M_Q = Q^6. \tag{13}$$

The nonzero vector field at the section is an exact eigenvector of Q with eigenvalue one. After an invertible, exactly stored dyadic similarity, the interval characteristic polynomial is divided by the exact factor $z - 1$. A Schur recursion certifies all 13 remaining roots of Q inside radius 0.7. Consequently the 13 nonflow multipliers of M_Q satisfy $|\mu| < 0.7^6 = 0.117649$.

Let B_a be the return of (7) over the relative interval L, starting from the certified section. The ordered full-period return for the first paired group is

$$M_1 = (B_2B_4B_1)^2. \tag{14}$$

The other two paired groups use the corresponding cyclic products. Each of these three enclosed cubic characteristic polynomials passes a Schur test at radius 0.13. At $N = 8$, these blocks supply nine stable directions. Together with the quotient, this proves the $(0, 1, 22)$ index and full-space local attraction. The spectrum in a restricted cluster subspace alone would not imply this conclusion.

4.3 The formerly absent blocks and their exact index

The two singleton blocks follow the alternating group path $3, 5, 3, 5, 3, 5$, so their full returns are

$$M_3 = (B_5B_3)^3, \qquad M_5 = (B_3B_5)^3. \tag{15}$$

These matrices are similar because B_3 is a fundamental matrix and is invertible. They therefore have the same spectrum. We nevertheless enclose and check both separately, retaining the uncertainty in every matrix product.

For each cubic $p(z) = z^3 + a_2z^2 + a_1z + a_0$, strict interval signs first give a real root outside the unit disk. Bisection yields the common outward-readable bracket

$$-23.083470 < \mu_u < -23.083469. \tag{16}$$

For an exact root u in this bracket, synthetic division gives

$$p(z) = (z - u)\left[z^2 + \alpha z + \beta\right], \quad \alpha = a_2 + u, \quad \beta = a_1 + u(a_2 + u). \tag{17}$$

All admissible coefficients and roots are enclosed when evaluating α and β. For a real monic quadratic, roots lie within radius R if

$$1 - |\beta|/R^2 > 0, \qquad 1 + \alpha/R + \beta/R^2 > 0, \qquad 1 - \alpha/R + \beta/R^2 > 0. \tag{18}$$

These strict inequalities hold at $R = 0.13$ for the entire enclosing coefficient box. Thus each singleton block has exactly one unstable root and two stable roots. A failed stability test alone would not give this classification; the root bracket and quadratic inequalities supply it.

In the replica, the paired groups provide $3(2k - 1)$ copies of stable three-dimensional blocks. The two formerly singleton groups provide $2(k-1)$ copies of blocks with index $(1, 0, 2)$. Including the quotient,

$$\begin{aligned} u &= 2(k - 1), \qquad n = 1, \\ s &= 13 + 9(2k - 1) + 4(k - 1) = 22k. \end{aligned}$$

This proves Theorem 1. All stable block bounds are below 0.13, so this same bound holds for every replicated stable multiplier.

Calculation	Bits	Steps/segment	Order	Verified conclusion
Cycle inclusion and spectrum	448	144	48	Attracting $N = 8$ cycle
Cycle recomputation	512	160	50	Same orbit and sharp bounds
Singleton integration	512	160	50	Real unstable roots
Singleton reintegration	640	192	54	Same enclosed spectrum
Spectral replay (both records)	768	—	—	Exact $(1, 0, 2)$ block index

Table 1. Distinct roles of the validated computations. Steps are counted within each of the three shooting segments. Spectral replay rereads the complete stored interval matrices and retains their radii.

Corollary 3 (Persistence near the certified parameters)**.** *There is an open neighborhood of* (ρ_*, r_*, b_*) *in which the relative five-cluster orbit continues and every balanced replica retains the index* (5)*.*

Proof. The verified nonsingularity of the shooting Jacobian and the implicit function theorem continue the relative orbit locally. The weights and permutation remain fixed. Its quotient and five contrast return matrices depend continuously on the parameters. All nonflow spectra are separated strictly from the unit circle. A common sufficiently small neighborhood therefore preserves the index of these finitely many blocks, and Proposition 2 gives the statement for every k. □

This corollary is qualitative; no numerical radius of the parameter neighborhood has been validated.

4.4 Recomputation and quantitative margins

Table 1 separates periodic-orbit validation, new variational integrations and spectral replay. The original periodic certificate and its recomputation use the same implementation and arithmetic library with different precision, subdivision and Taylor settings. They are not independent proof implementations. Independent field/Jacobian embeddings and exact rational Taylor-coefficient controls supplement them.

For the 512-bit cycle recomputation, the Krawczyk contraction bound is below 1.960×10^{-19} and the maximum component inclusion ratio is below 2.574×10^{-12}. The shooting-box radius is 10^{-38}. The 640-bit singleton reintegration also passes the independent embedding check against the full 47-dimensional $N = 16$ Jacobian, including the group containing the eliminated displacement coordinate. There are 1,692 checked components at rational test states. These algebra controls detect implementation inconsistencies; the invariant splitting itself is derived in Section 3.

The quadratic radius-0.13 inequalities remain well separated from zero; their smallest interval lower margin exceeds 0.9999. Source snapshots, input hashes, interval matrices and the separate spectral replay are preserved for both integration settings. The computational proof remains conditional on the correctness of the stated algorithms, implementation and arithmetic library; it is not proof-assistant formalization.

5 Nonlinear numerical illustrations

The figures use ordinary floating-point integrations of the full $3N - 1$ dimensional equations. They provide geometric and dynamical interpretation of the certified linear stability. They are not used to prove the existence of the periodic orbit or its Floquet classification.

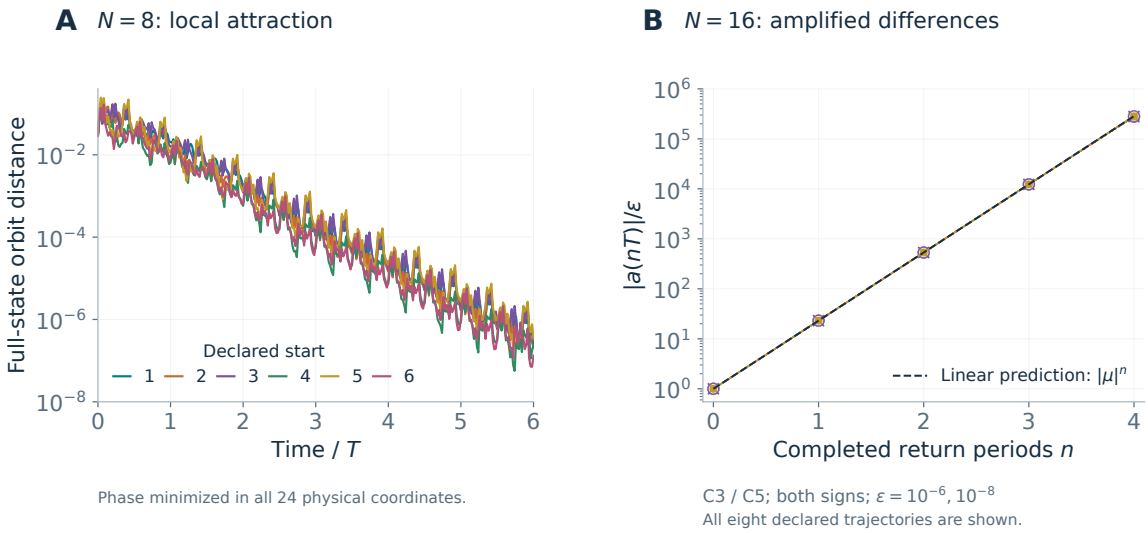


Figure 2. Full-system numerical checks of the two stability outcomes. (A) Every declared nearby $N = 8$ trajectory, with numerical phase minimization in the full physical state. (B) All eight $N = 16$ starts: two newly available contrast blocks, two signs and two perturbation norms. The signed contrast component a is sampled at nT; absolute values are normalized by the initial norm ε. Solid and dotted curves correspond to 10^{-8} and 10^{-6}, respectively; circles and crosses distinguish signs. The dashed line is the linear Floquet prediction. Overlapping curves are retained.

5.1 Approach to the eight-row orbit

Six starts were declared at phases $jT/6$, $j = 0, \ldots, 5$, with random perturbations of full physical-state norm 0.03. The displacement perturbations have zero sum, and all three paired groups are initially split. The fixed random seed and all initial states are retained. No clustering condition is imposed during the integration.

Distance is measured in all 24 physical coordinates, including the reconstructed displacement, and minimized numerically over the phase of the reference orbit. After six periods, all six distances fall from approximately 0.03 to between 1.32×10^{-7} and 3.42×10^{-7} (Fig. 2A). Some trajectories first move farther from the orbit, so local attraction does not imply monotone decay in this norm. Tightening the DOP853 settings changes any displayed state by less than 8.62×10^{-11}.

These six local examples do not estimate a basin volume or a certified basin radius. The repository also retains an earlier declared amplitude 0.6 set: none of those six starts passed the six-period approach gate, and five failed long-horizon pointwise solver agreement. Their eventual destinations are not classified. All amplitudes remain in the data audit.

5.2 Sign-reversing departure at sixteen rows

We replicate the certified section state to groups of sizes $(4, 4, 2, 4, 2)$. For each formerly singleton group, a numerical eigenvector v of the midpoint return matrix proposes an unstable direction. Its two rows receive perturbations $\pm\varepsilon v/\sqrt{2}$, so the full physical perturbation has norm ε and leaves all group means unchanged at the initial instant. Both overall signs and $\varepsilon \in \{10^{-6}, 10^{-8}\}$ give eight declared starts. The full nonlinear dynamics are integrated for four periods.

For the normalized row contrast $\xi = (\delta q_a - \delta q_b)/\sqrt{2}$, let $a = v^{\mathsf{T}}\xi$. At small amplitude the stroboscopic prediction is $a(nT) \simeq a(0)\mu_u^n$. The negative multiplier produces alternation of sign, while its magnitude gives rapid amplification (Figs. 2B and 3C). For all four 10^{-8} starts, relative discrepancies from this prediction at the first three returns are below 10^{-6}. The larger-amplitude

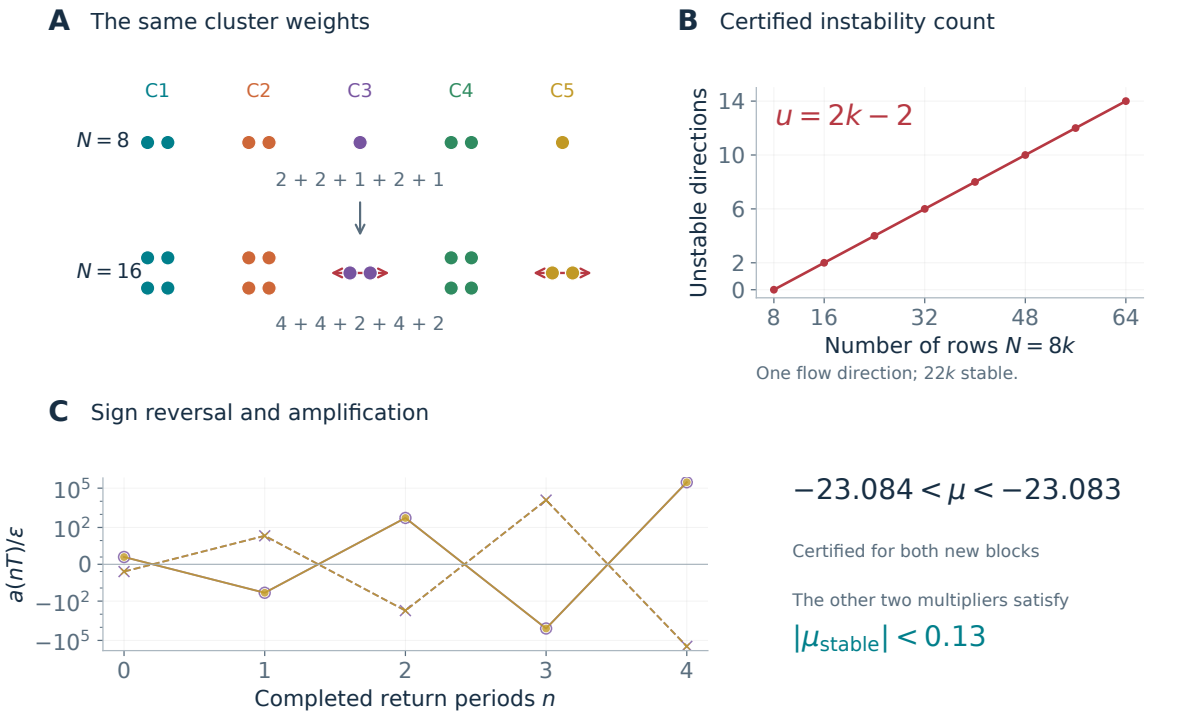


Figure 3. The mechanism and its quantitative consequence. (A) Enlarging all groups preserves their weights. Red arrows mark new internal differences in the two original singletons. (B) The exact unstable count $2k-2$; displayed points are discrete system sizes, and the theorem holds for every integer $k \geq 1$. (C) Both signs in both new blocks at $\varepsilon = 10^{-8}$, shown on a symmetric logarithmic vertical scale. The displayed multiplier bounds are interval-certified; the return trajectories are numerical illustrations.

starts also remain in the archived ensemble when nonlinear corrections become visible.

Both DOP853 settings are retained for every trajectory. Their maximum pointwise state difference is below 2.69×10^{-8}. A separate Radau integration, using the analytic full Jacobian, checks one complete larger-amplitude history and differs by less than 1.97×10^{-8}. These are finite numerical controls and do not identify the eventual attractors reached after departure.

6 Discussion

The five-group state supplies a concrete distinction between preserving a collective motion and preserving its attraction. The quotient orbit does not change under balanced replication, yet the admissible tangent space does. Two previously absent blocks become available, and their unstable multiplier fixes the complete growth in saddle index. Existing cluster-stability theory provides the organizing framework; the explicit periodic orbit, interval enclosures and exact index formula establish the model-specific result.

The bounds also show that the effect is robust to sufficiently small parameter changes. It does not rely on a multiplier lying close to one or on a long transient whose classification is uncertain. Quantifying the parameter neighborhood and locating the nonlinear destinations of departing replicas are natural extensions.

Separate computations in this model exhibit a candidate regime with six positive finite-time Lyapunov estimates and very long switching trajectories. Those results are documented in the repository, but do not enter the theorem proved here. A connection from the certified cycle or its replicas to those regimes would require additional global analysis. The present paper establishes

a periodic-attractor and periodic-saddle result, without an asymptotic chaos claim or a claim of historical priority for cluster desynchronization as a general mechanism.

Data and code availability

The complete materials are available in the `chaos-attractor-lab` repository. The manuscript directory is `papers/replication-instability`. Its README maps claims and figures to immutable evidence records and gives regeneration commands. The article manifest supplies raw-byte SHA-256 hashes. The certificate inputs use lossless dyadic midpoint/radius serialization. All numerical initial conditions, settings and outcomes are retained. The figures are available as vector PDF and SVG, with PNG previews and a rotating view whose camera motion is explicitly distinguished from dynamical evolution.

A Density interpretation and physical admissibility

Let ℓ be the spatial period, $a_0 > 0$ the constant total transition rate, κ the elastic stiffness, η the drag coefficient and U the potential amplitude. With dimensional density Q_j and relative shift Δ_j, the normalization is

$$\theta = \frac{2\pi\xi}{\ell}, \quad \tau = a_0 t, \quad q_j = \ell Q_j, \quad X_j = \frac{2\pi\Delta_j}{\ell}, \quad \rho = \frac{\kappa}{\eta a_0}, \quad \gamma = \frac{2\pi^2 U}{\eta \ell^2 a_0}.$$

For transition-rate Fourier coefficients a_1, b_1, put $A = a_1/a_0$ and $B = b_1/a_0$. Under a cosine interaction potential, constant total transition rate and sinusoidal forcing, write the normalized density equation as

$$\partial_\tau q_j + V_j \partial_\theta q_j = -q_j + \frac{1}{2} + A\cos\theta + B\sin\theta. \tag{19}$$

Setting the first Fourier coefficients to Ac_j and As_j, with $b = B/A$ and $r = -\gamma A$, gives (2). Only the first sine coefficient enters the force. Thus this finite system is an exact closure under the stated assumptions. Higher-mode relaxation is already present in the source model [1].

For $b = 1$, the row amplitude satisfies

$$(c_j^2 + s_j^2)' = -2(c_j^2 + s_j^2) + 2(c_j + s_j).$$

At a maximum along a periodic solution this implies $c_j^2 + s_j^2 \le 2$. Hence the lifted density $q_j = 1/2 + A(c_j\cos\theta + s_j\sin\theta)$ is positive whenever $|A|\sqrt{2} < 1/2$. The source activity-path convention satisfies this condition at the parameter used here. This statement concerns the specified density model; a mechanically bending flagellum coupled to fluid would require additional state variables and analysis.

An omitted Fourier coefficient of order $n \ge 2$ evolves as

$$z_{j,n}(\tau) = e^{-\tau + in[X_j(\tau) - X_j(0)]} z_{j,n}(0).$$

The omitted mean relaxes at the same exponential rate. Since $X_j(T) = X_j(0)$, these residual modes have return multiplier e^{-T} and add only stable directions. More directly, the residual density is an exponentially damped spatial translation; its periodic Sobolev norm is multiplied exactly by $e^{-\tau}$. This proves local attraction of the lifted $N = 8$ cycle in these norms. The replicas' finite-dimensional unstable directions persist within the invariant zero-tail subspace.

B A compact verification recipe

The decisive records, all relative to the repository root, are

1. `results/motor-five-cycle-proof/recheck-01.json`: periodic inclusion, quotient and paired spectra at 512 bits;
2. `results/motor-replication-article/reintegration-640.json`: new singleton return matrices from a 640-bit variational integration;
3. `results/motor-replication-article/exact-index-recheck.json`: strict real-root brackets and quadratic radius tests at 768 bits;
4. `results/motor-replication-article/nonlinear/`: the declared eight-start illustration and a different-solver control.

The new index checker recomputes each cubic from the stored interval matrix, cross-checks the coefficients with explicit 3×3 formulas, brackets the negative root by strict signs and verifies (18). It never substitutes a midpoint eigenvalue for an interval statement. A second replay uses the earlier 512-bit matrices. Reproduction writes to unused output paths and preserves all earlier records. The README supplies the separate proof and visualization dependency sets; the proof requires no GPU, NumPy or SciPy.

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